Trying to predict a superbowl winner using NFL stats from the 2025 season

Step 1: adding a win column to 2025 dataset.

```
 other_dataset <- load_schedules(2025) %>%
  filter(week <= 18) %>%
  select(season, week, home_team, away_team, home_score, away_score)

 win_condition <- other_dataset %>%
   pivot_longer(
    cols = c(home_team, away_team),
    names_to = "home_away",
    values_to = "team"
  ) %>%
  mutate(
    win = case_when(
      home_away == "home_team" & home_score > away_score ~ 1,
      home_away == "away_team" & away_score > home_score ~ 1,
      TRUE ~ 0
    )
  ) %>%
  select(season, week, team, win)


NFL_full_stats <- load_team_stats(2025) %>%
  left_join(win_condition, by = c("season", "week", "team")) %>%
  relocate(win, .after = "team")
```

Step 2: creating a dataset strictly with 2025 nfl playoff teams and optimizing variables

```
NFL_2025_team_stats <- NFL_full_stats %>%
  filter(season_type == "REG") %>%
  mutate(comp_perc = completions/attempts,
  turnovers = passing_interceptions + sack_fumbles_lost + rushing_fumbles_lost + receiving_fumbles_lost
  forced_turnovers = def_interceptions + fumble_recovery_opp,
  rypa = rushing_yards / carries, ##rypa = rushing yards per attempt
  total_tackles = def_tackles_solo + def_tackles_with_assist + def_tackle_assists) %>%
  select(season, week, team, opponent_team, comp_perc, passing_yards, passing_tds, sacks_suffered,

playoffteams <- c("BUF", "CAR", "SEA", "LA", "LAC", "PIT", "NE", "DEN", "JAX", "HOU", "SF", "GB", "PHI"

NFL_2025_playoff_teams <- NFL_2025_team_stats %>%
  filter(team %in% playoffteams)
```

I took the playoff teams of 2025 and filtered all of their 17 games of the season while adding parameters that I think will be usefully in predicting playoff success.

Step 3: only including the variables I think are the most important to winning and losing football games, while also finding average of this variables from all 14 playoff teams to use as a benchmark later on.

```
avg_benchmark_playoff_teams <- NFL_2025_playoff_teams %>%
  select(-"season", -"week") %>%
  summarise(across(where(is.numeric), mean, na.rm = TRUE))
```

```r
avg_benchmark_teams <- NFL_2025_team_stats %>%
  select(-"season", -"week") %>%
  summarise(across(where(is.numeric), mean, na.rm = TRUE))
```

I selected all of the parameters I think will be the most impactful so that I can average them over the season and starting using statistical methods.

Step 4: Finding the average on every variable for each specific team over the 2025 season.

```r
per_by_playoff_team_average <- NFL_2025_playoff_teams %>%
  group_by(team) %>%
  select(-season, -week) %>%
  summarise(across(where(is.numeric), mean, na.rm = TRUE))

per_by_team_average <- NFL_2025_team_stats  %>%
  group_by(team) %>%
  select(-season, -week) %>%
  summarise(across(where(is.numeric), mean, na.rm = TRUE))
```

this gives me season averages of every playoff team

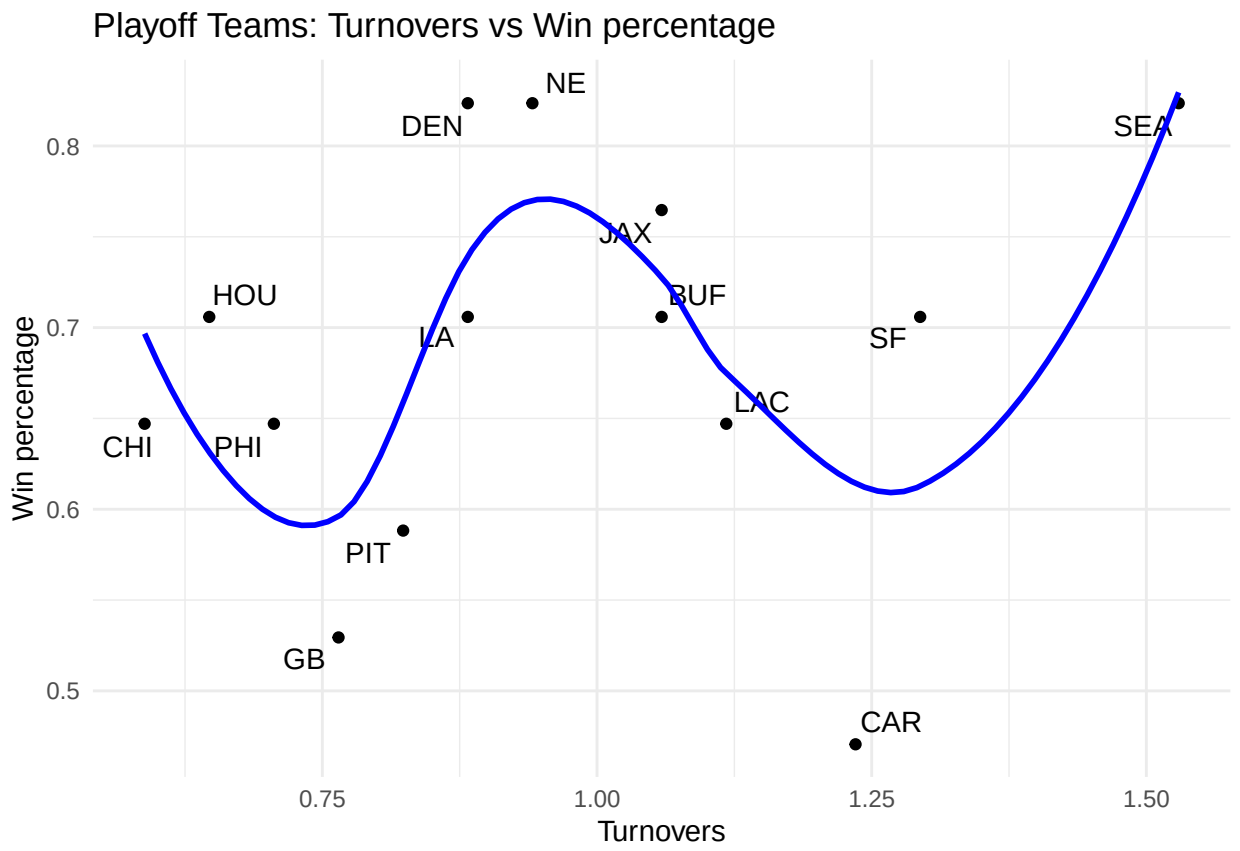
Step 5: Turnover vs Win and Forced turnovers vs Win graphs for Playoff teams and rest of league.

```r
turnover_to_win_playoff <- ggplot(per_by_playoff_team_average, aes(x = turnovers, y = win, label = team
  geom_point() +
  geom_text_repel() +
  geom_smooth(se = FALSE, color = "blue") +
  theme_minimal() +
  labs(title = "Playoff Teams: Turnovers vs Win percentage",
       x = "Turnovers",
       y = "Win percentage")

turnover_to_win <- ggplot(per_by_team_average, aes(x = turnovers, y = win, label = team)) +
  geom_point() +
  geom_text_repel() +
  geom_smooth(se = FALSE, color = "blue") +
  theme_minimal() +
  labs(title = "ALL NFL Teams: Turnovers vs Win percentage",
       x = "Turnovers",
       y = "Win percentage")

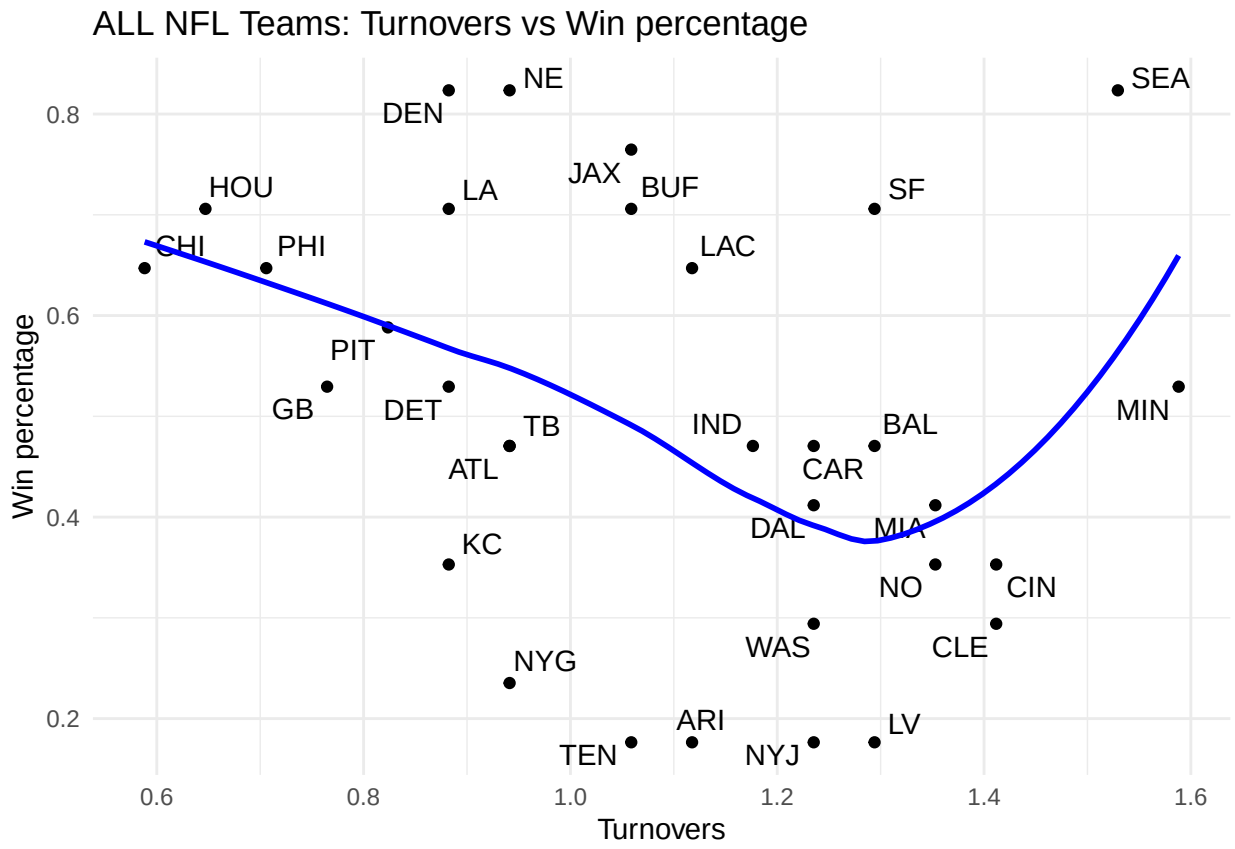
turnover_to_win_playoff
```

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```




```
turnover_to_win
```

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

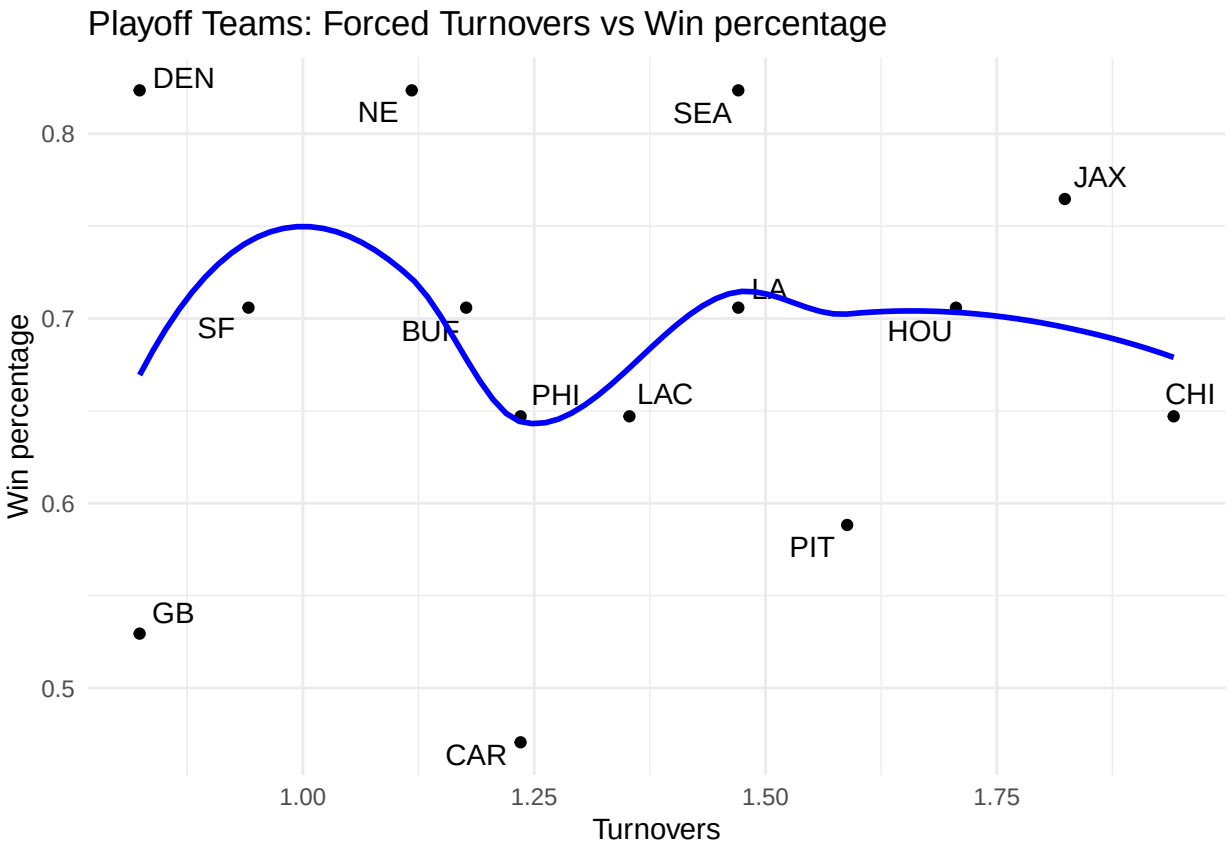



```
forced_turnover_to_win_playoff <- ggplot(per_by_playoff_team_average, aes(x = forced_turnovers, y = win
  geom_point() +
  geom_text_repel() +
  geom_smooth(se = FALSE, color = "blue") +
  theme_minimal() +
  labs(title = "Playoff Teams: Forced Turnovers vs Win percentage",
       x = "Turnovers",
       y = "Win percentage")

forced_turnover_to_win <- ggplot(per_by_team_average, aes(x = forced_turnovers, y = win, label = team))
  geom_point() +
  geom_text_repel() +
  geom_smooth(se = FALSE, color = "blue") +
  theme_minimal() +
  labs(title = "All NFL Teams: Forced Turnovers vs Win percentage",
       x = "Turnovers",
       y = "Win percentage")

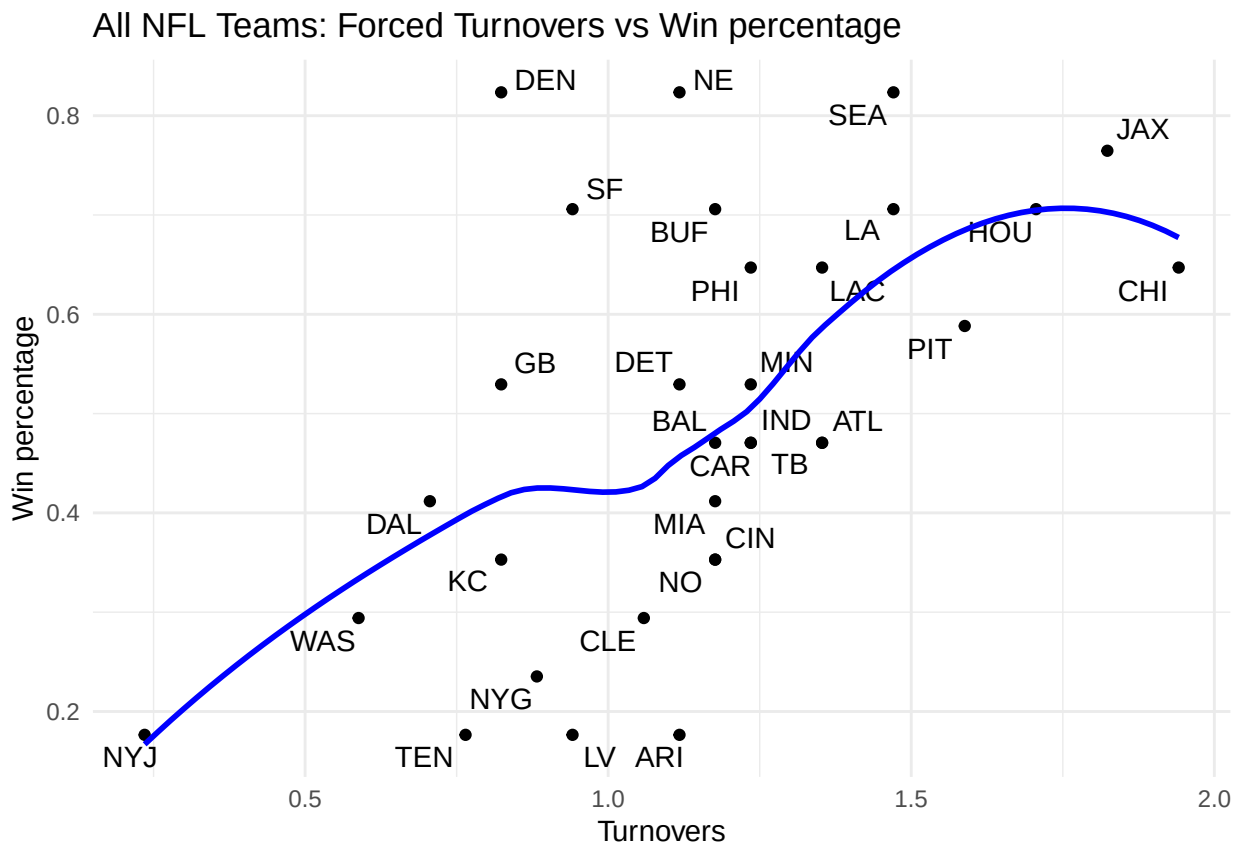
forced_turnover_to_win_playoff
```

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```




```
forced_turnover_to_win
```

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



Step 6: more graphs

```
playoff_Tackles_for_loss <- ggplot(per_by_playoff_team_average, aes(x = def_tackles_for_loss, y = win, l
  geom_point() +
  geom_text_repel() +
  geom_smooth(se = FALSE, color = "blue") +
  theme_minimal() +
  labs(title = "Playoff Teams: Tackles for loss vs Win percentage",
       x = "TFL",
       y = "Win percentage")

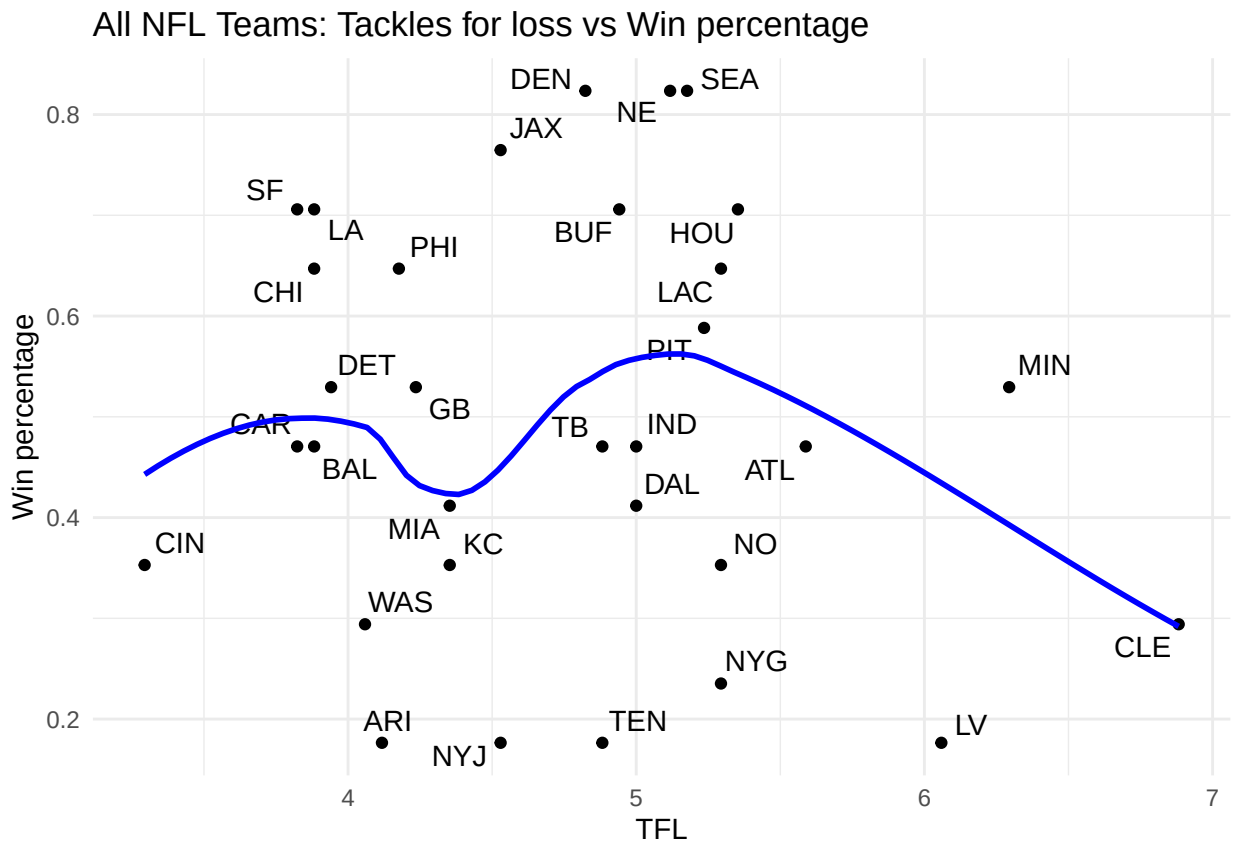
Tackles_for_loss <- ggplot(per_by_team_average, aes(x = def_tackles_for_loss, y = win, label = team)) +
  geom_point() +
  geom_text_repel() +
  geom_smooth(se = FALSE, color = "blue") +
  theme_minimal() +
  labs(title = "All NFL Teams: Tackles for loss vs Win percentage",
       x = "TFL",
       y = "Win percentage")

Tackles_for_loss
```

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'

## Warning: The following aesthetics were dropped during statistical transformation: label.
## i This can happen when ggplot fails to infer the correct grouping structure in
```

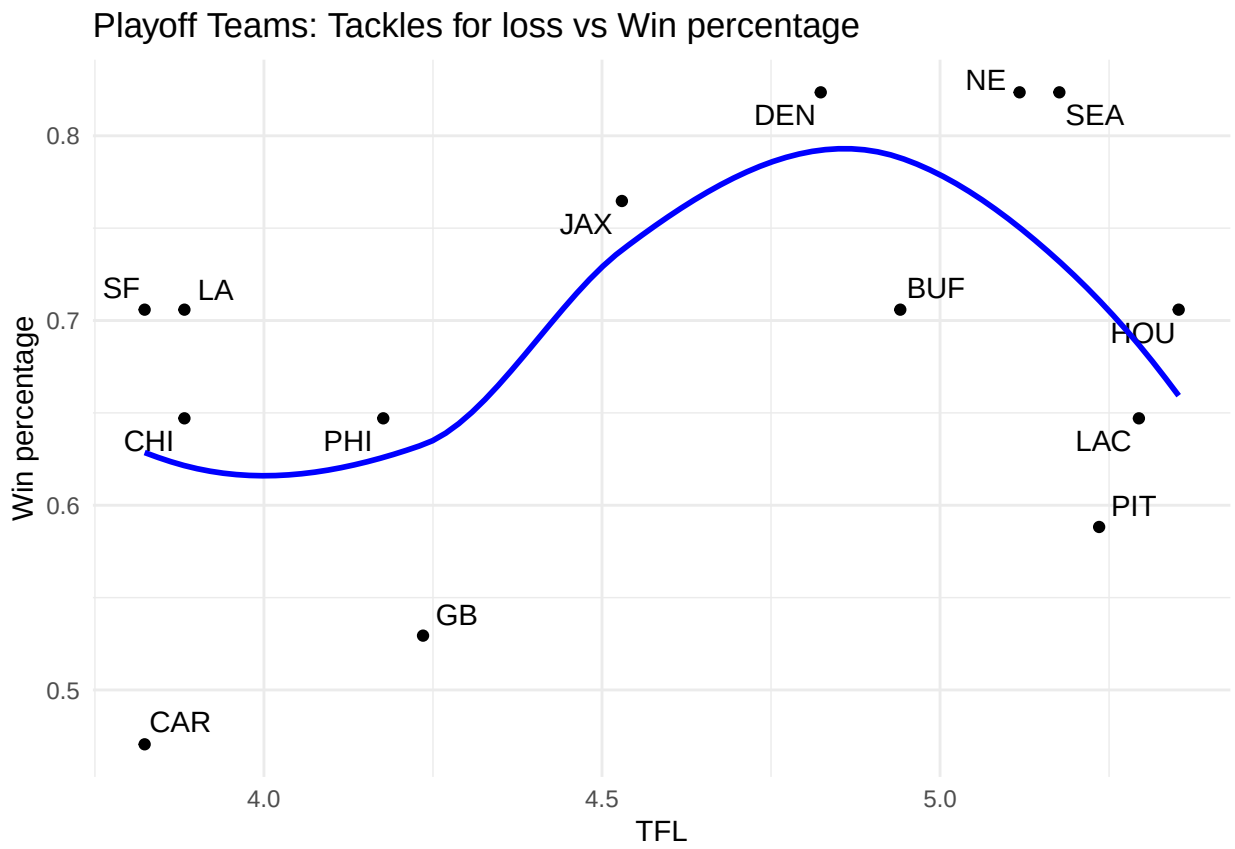
```
##   the data.
## i Did you forget to specify a ‘group‘ aesthetic or to convert a numerical
##   variable into a factor?
```




```
playoff_Tackles_for_loss
```

```
## ‘geom_smooth()‘ using method = ’loess’ and formula = ’y ~ x’
```

```
## Warning: The following aesthetics were dropped during statistical transformation: label.
## i This can happen when ggplot fails to infer the correct grouping structure in
##   the data.
## i Did you forget to specify a ‘group‘ aesthetic or to convert a numerical
##   variable into a factor?
```




```
playoff_penalties <- ggplot(per_by_playoff_team_average, aes(x = penalties, y = win, label = team)) +
  geom_point() +
  geom_text_repel() +
  geom_smooth(se = FALSE, color = "blue") +
  theme_minimal() +
  labs(title = "Playoff Teams: Penalties vs Win percentage",
       x = "Penalties",
       y = "Win percentage")

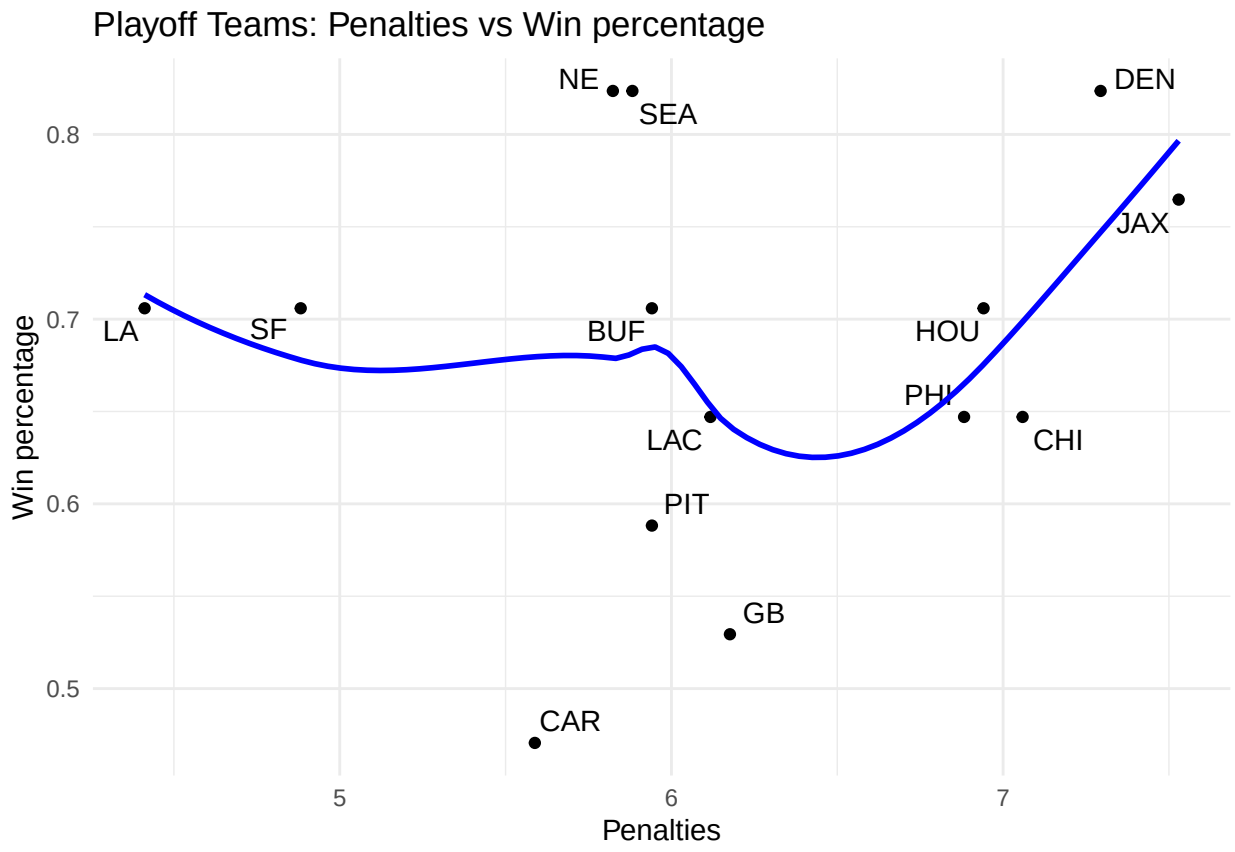
penalties <- ggplot(per_by_team_average, aes(x = penalties, y = win, label = team)) +
  geom_point() +
  geom_text_repel() +
  geom_smooth(se = FALSE, color = "blue") +
  theme_minimal() +
  labs(title = "All NFL Teams: Penalties vs Win percentage",
       x = "Penalties",
       y = "Win percentage")

playoff_penalties
```

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

```
## Warning: The following aesthetics were dropped during statistical transformation: label.
## i This can happen when ggplot fails to infer the correct grouping structure in
##   the data.
```

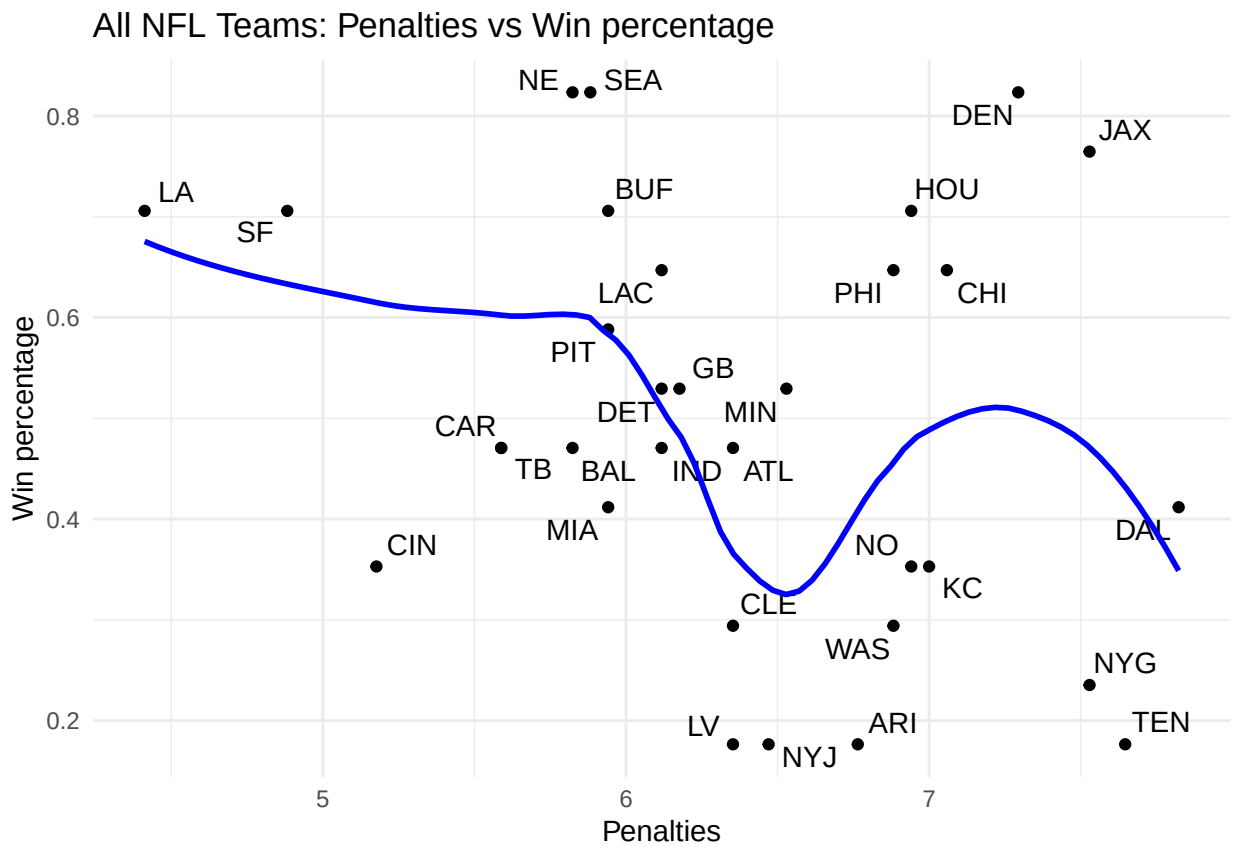
```
## i Did you forget to specify a ‘group‘ aesthetic or to convert a numerical
##   variable into a factor?
```




```
penalties
```

```
## ‘geom_smooth()‘ using method = ’loess’ and formula = ’y ~ x’
```

```
## Warning: The following aesthetics were dropped during statistical transformation: label.
## i This can happen when ggplot fails to infer the correct grouping structure in
##   the data.
## i Did you forget to specify a ‘group‘ aesthetic or to convert a numerical
##   variable into a factor?
```



Step 7: Creating a strength variable to use for the modeling

```
logit_model_1 <- glm(
  win ~ sacks_suffered + turnovers + forced_turnovers +  rushing_epa + passing_epa + def_tackles_for_los
  data = NFL_2025_playoff_teams,
  family = binomial
)

summary(logit_model_1)
```

```
##
## Call:
## glm(formula = win ~ sacks_suffered + turnovers + forced_turnovers +
##     rushing_epa + passing_epa + def_tackles_for_loss + def_sacks +
##     penalties, family = binomial, data = NFL_2025_playoff_teams)
##
## Coefficients:
##                      Estimate Std. Error z value Pr(>|z|)
## (Intercept)           0.18794    0.71543   0.263 0.792792
## sacks_suffered        0.05199    0.12557   0.414 0.678880
## turnovers            -0.71723    0.20449  -3.507 0.000453 ***
## forced_turnovers      0.85346    0.19353   4.410 1.03e-05 ***
## rushing_epa           0.09025    0.03641   2.479 0.013189 *
## passing_epa           0.12108    0.02574   4.704 2.55e-06 ***
## def_tackles_for_loss  0.08278    0.10963   0.755 0.450198
## def_sacks             0.39635    0.14356   2.761 0.005765 **
```

```
## penalties            -0.18225    0.06998  -2.604 0.009205 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 296.62  on 237  degrees of freedom
## Residual deviance: 184.24  on 229  degrees of freedom
## AIC: 202.24
##
## Number of Fisher Scoring iterations: 6
```

```
NFL_2025_playoff_teams$win_pct <- NFL_2025_playoff_teams$win

NFL_2025_playoff_teams$win_pct <- predict(logit_model_1, type = "response")


team_strength <- NFL_2025_playoff_teams %>%
  group_by(team) %>%
  summarise(
    strength = mean(win_pct, na.rm = TRUE),
    win_pct = mean(win),
    wins = sum(win)
  ) %>%
  arrange(desc(strength))
```

Step 8: Creating the sim and running the wild card round to determine the divisional round

```
sim_game <- function(team_a, team_b, team_strength) {
  prob_a = team_strength$strength[team_strength$team == team_a]
  prob_b = team_strength$strength[team_strength$team == team_b]

  win_prob_a = ((prob_a) - (prob_a * prob_b))/((prob_a + prob_b) - (2 * (prob_a * prob_b)))

  if (runif(1) < win_prob_a) return(team_a) else return(team_b)
}

  runsim_wc <- function(team_strength) {
    afc_wc1 <- sim_game("NE", "LAC", team_strength)
    afc_wc2 <- sim_game("JAX", "BUF", team_strength)
    afc_wc3 <- sim_game("HOU", "PIT", team_strength)

    nfc_wc1 <- sim_game("CHI", "GB", team_strength)
    nfc_wc2 <- sim_game("PHI", "SF", team_strength)
    nfc_wc3 <- sim_game("LA", "CAR", team_strength)


    return(c(afc_wc1, afc_wc2, afc_wc3, nfc_wc1, nfc_wc2, nfc_wc3))
  }

set.seed(123)
wc_results <- replicate(10000, runsim_wc(team_strength))
```

```r
all_wc_winners <- as.vector(wc_results)

wc_odds <- as.data.frame(table(all_wc_winners)) %>%
  rename(team = all_wc_winners, wins = Freq) %>%
  mutate(win_prob = wins / 10000) %>%
  arrange(desc(win_prob))
```

Step 9: running the divisional round to determine the championship round

```r
runsim_dv <- function(team_strength) {
    afc_dv1 <- sim_game("NE", "PIT", team_strength)
    afc_dv2 <- sim_game("DEN", "BUF", team_strength)

    nfc_dv1 <- sim_game("CHI", "PHI", team_strength)
    nfc_dv2 <- sim_game("SEA", "LA", team_strength)


    return(c(afc_dv1, afc_dv2, nfc_dv1, nfc_dv2))
  }

set.seed(123)
dv_results <- replicate(10000, runsim_dv(team_strength))

all_dv_winners <- as.vector(dv_results)

dv_odds <- as.data.frame(table(all_dv_winners)) %>%
  rename(team = all_dv_winners, wins = Freq) %>%
  mutate(win_prob = wins / 10000) %>%
  arrange(desc(win_prob))
```

Step 10: running the championship games to determine the superbowl

```r
runsim_ch <- function(team_strength) {
    afc_ch <- sim_game("NE", "DEN", team_strength)
    nfc_ch <- sim_game("LA", "CHI", team_strength)

    return(c(afc_ch, nfc_ch))
  }

set.seed(123)
ch_results <- replicate(10000, runsim_ch(team_strength))

all_ch_winners <- as.vector(ch_results)

ch_odds <- as.data.frame(table(all_ch_winners)) %>%
  rename(team = all_ch_winners, wins = Freq) %>%
  mutate(win_prob = wins / 10000) %>%
  arrange(desc(win_prob))
```

```r
runsim_sb <- function(team_strength) {
    sb <- sim_game("NE", "LA", team_strength)
```

```
    return(c(sb))
  }

set.seed(123)
sb_results <- replicate(10000, runsim_sb(team_strength))

all_sb_winners <- as.vector(sb_results)

sb_odds <- as.data.frame(table(all_sb_winners)) %>%
  rename(team = all_sb_winners, wins = Freq) %>%
  mutate(win_prob = wins / 10000) %>%
  arrange(desc(win_prob))
```

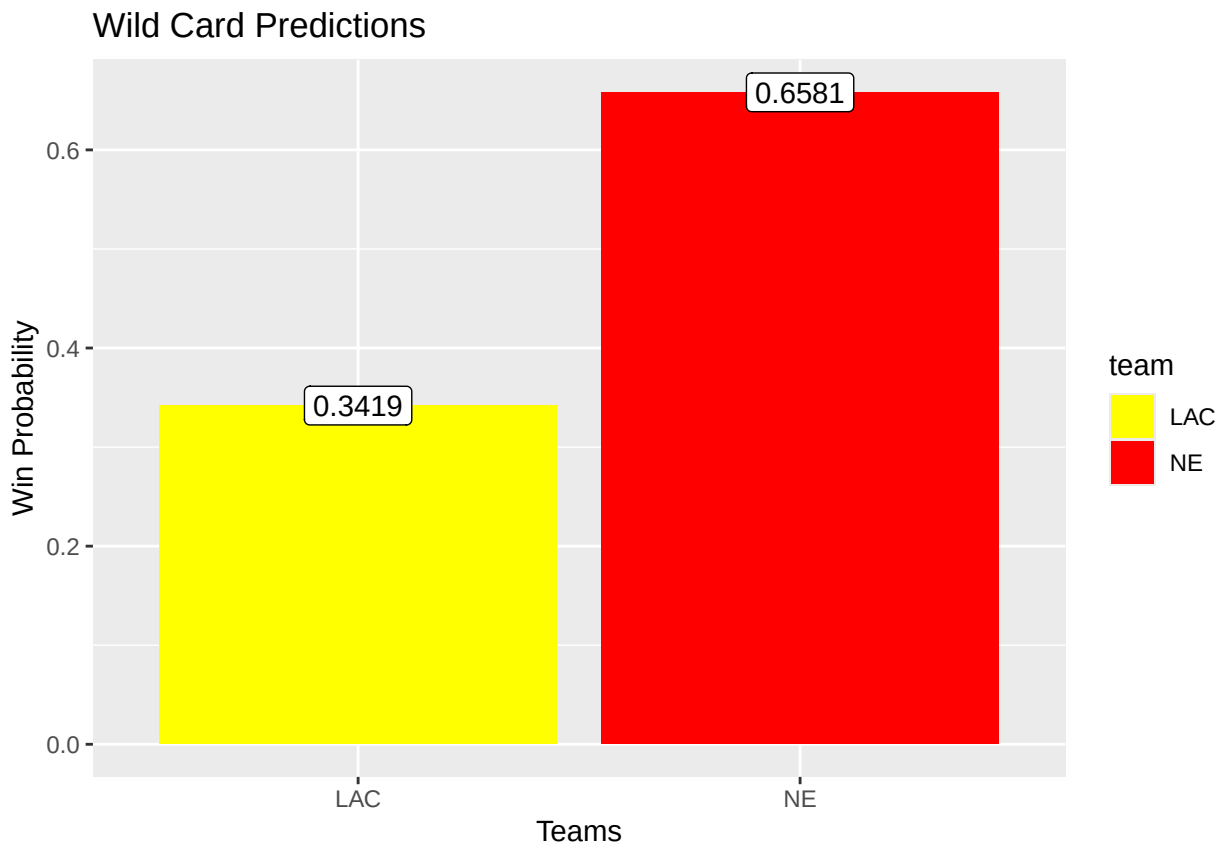
Step 11: GRAPHS

```
NE_LAC <- wc_odds %>%
  filter(team %in% c("NE", "LAC"))

afc_wc1_chart <- ggplot(NE_LAC, aes(x = team, y = win_prob, fill = team)) +
  geom_col() +
  geom_label(aes(label = win_prob, accuracy = 0.1),color = "black", fill = "white") +
  scale_fill_manual(values = c("LAC" = "yellow", "NE" = "red")) +
  labs(title = "Wild Card Predictions",
       y = "Win Probability",
       x = "Teams")
```

```
## Warning in geom_label(aes(label = win_prob, accuracy = 0.1), color = "black", :
## Ignoring unknown aesthetics: accuracy
```

```
afc_wc1_chart
```

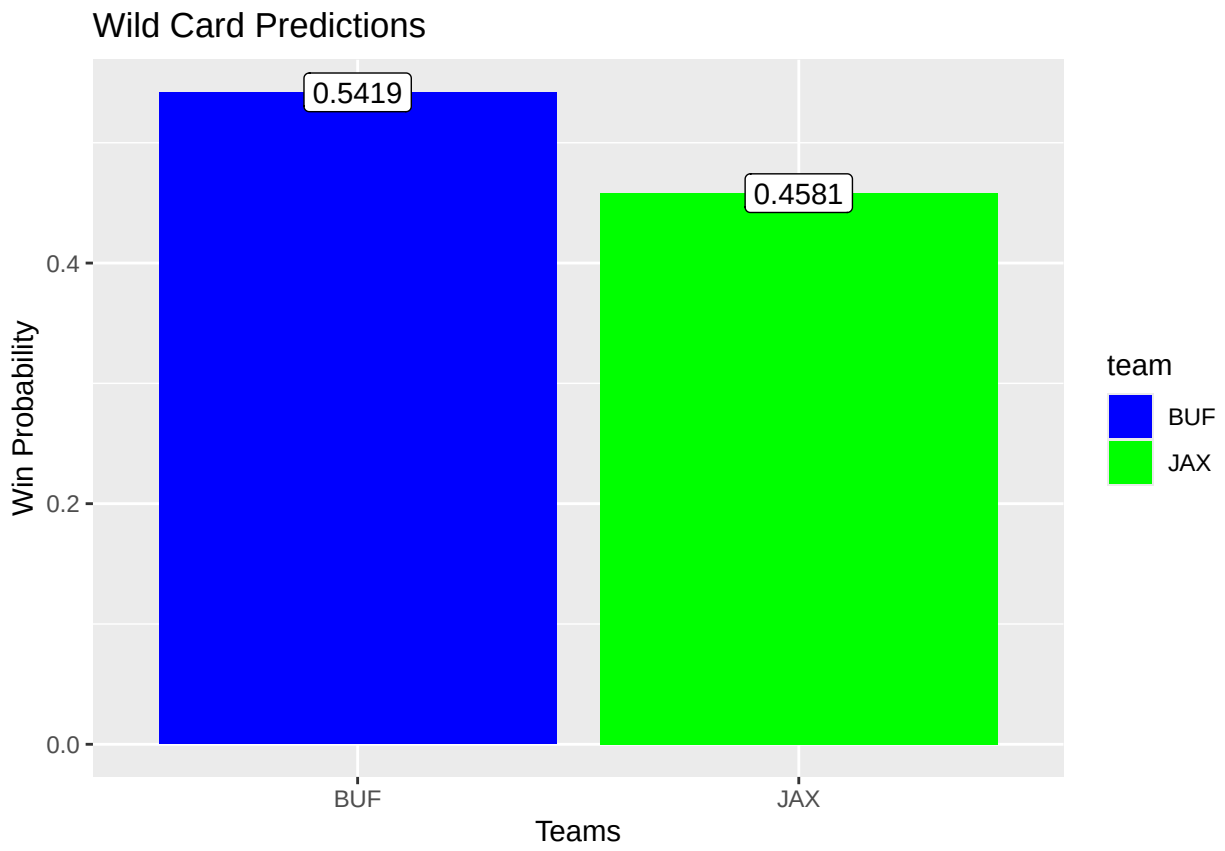



```
JAX_BUF <- wc_odds %>%
  filter(team %in% c("JAX", "BUF"))

afc_wc2_chart <- ggplot(JAX_BUF, aes(x = team, y = win_prob, fill = team)) +
  geom_col() +
  geom_label(aes(label = win_prob, accuracy = 0.1),color = "black", fill = "white") +
  scale_fill_manual(values = c("JAX" = "green", "BUF" = "blue")) +
  labs(title = "Wild Card Predictions",
       y = "Win Probability",
       x = "Teams")
```

```
## Warning in geom_label(aes(label = win_prob, accuracy = 0.1), color = "black", :
## Ignoring unknown aesthetics: accuracy
```

```
afc_wc2_chart
```

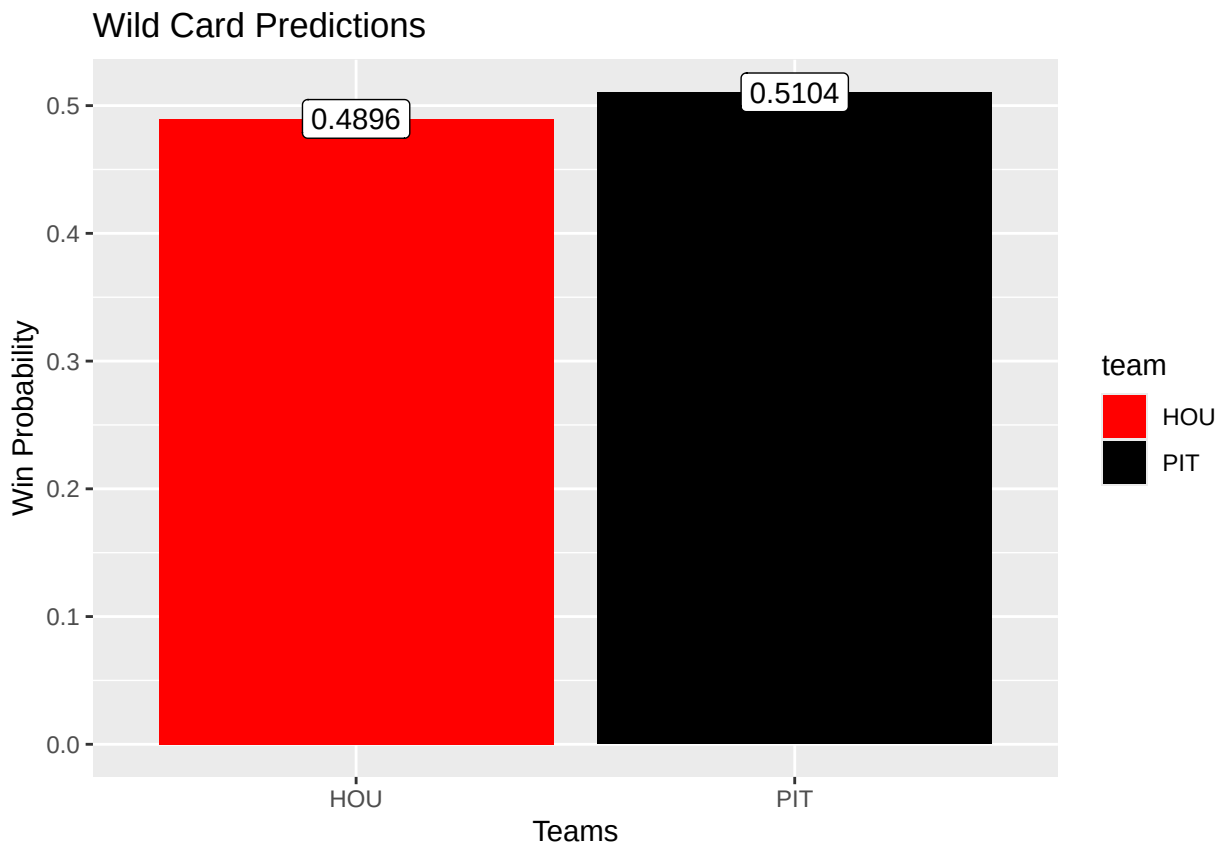



```
PIT_HOU <- wc_odds %>%
  filter(team %in% c("PIT", "HOU"))

afc_wc3_chart <- ggplot(PIT_HOU, aes(x = team, y = win_prob, fill = team)) +
  geom_col() +
  geom_label(aes(label = win_prob, accuracy = 0.1),color = "black", fill = "white") +
  scale_fill_manual(values = c("PIT" = "black", "HOU" = "red")) +
  labs(title = "Wild Card Predictions",
       y = "Win Probability",
       x = "Teams")
```

```
## Warning in geom_label(aes(label = win_prob, accuracy = 0.1), color = "black", :
## Ignoring unknown aesthetics: accuracy
```

```
afc_wc3_chart
```

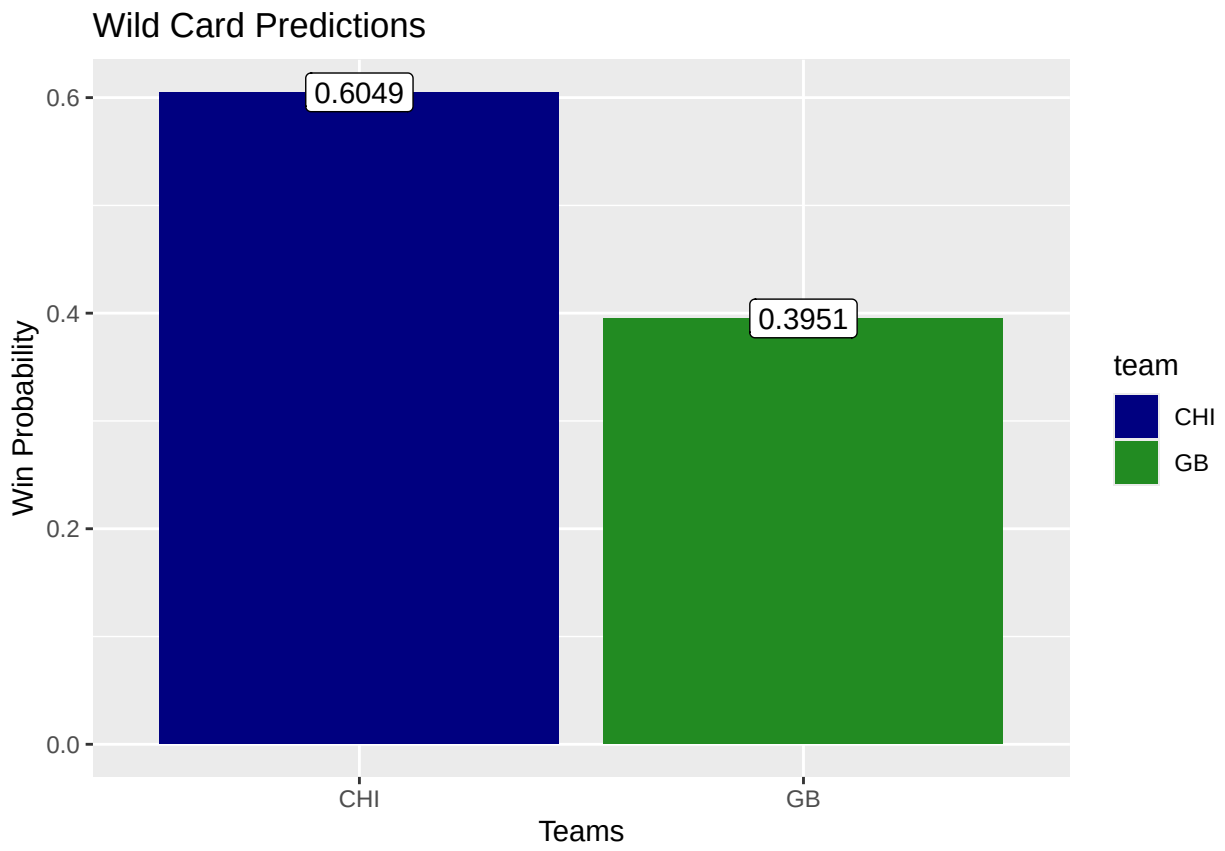



```
CHI_GB <- wc_odds %>%
  filter(team %in% c("CHI", "GB"))

nfc_wc1_chart <- ggplot(CHI_GB, aes(x = team, y = win_prob, fill = team)) +
  geom_col() +
  geom_label(aes(label = win_prob, accuracy = 0.1),color = "black", fill = "white") +
  scale_fill_manual(values = c("CHI" = "navy", "GB" = "forestgreen")) +
  labs(title = "Wild Card Predictions",
       y = "Win Probability",
       x = "Teams")
```

```
## Warning in geom_label(aes(label = win_prob, accuracy = 0.1), color = "black", :
## Ignoring unknown aesthetics: accuracy
```

```
nfc_wc1_chart
```

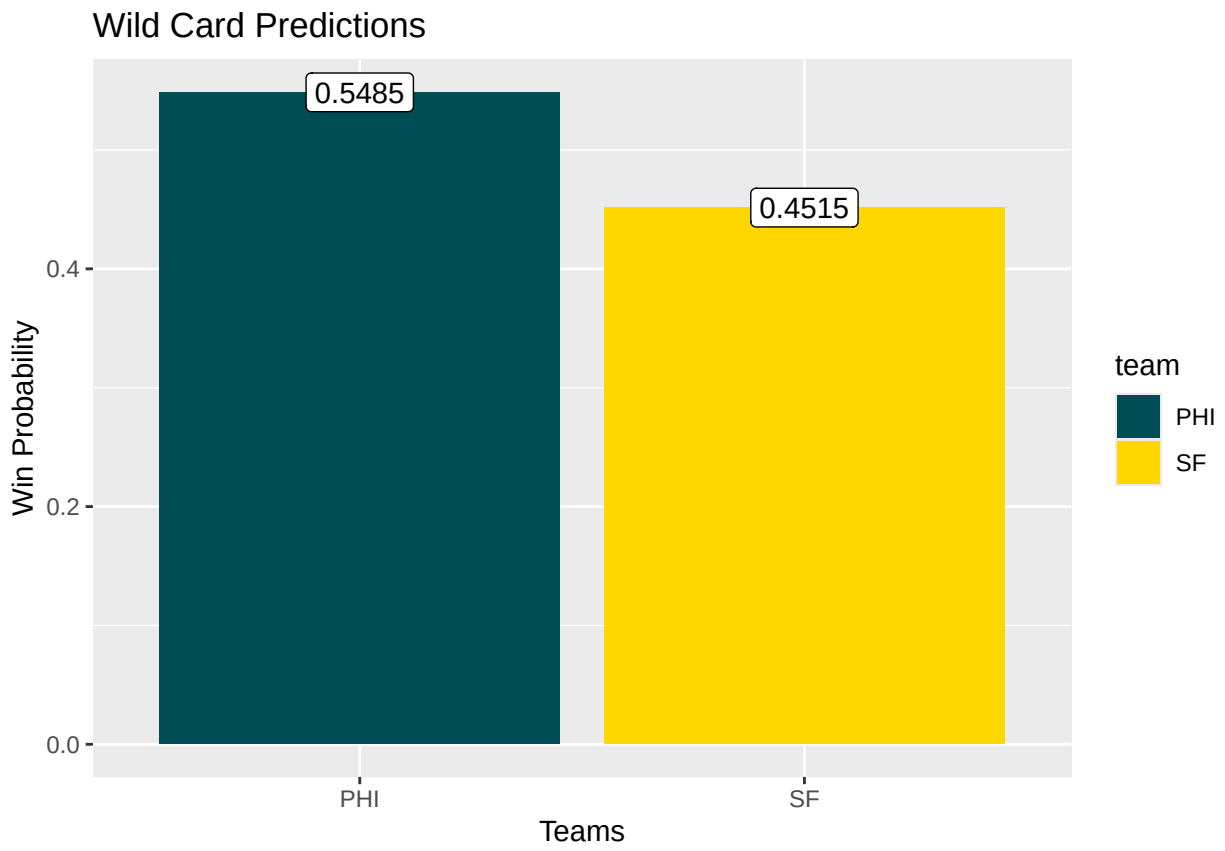



```
PHI_SF <- wc_odds %>%
  filter(team %in% c("PHI", "SF"))

nfc_wc2_chart <- ggplot(PHI_SF, aes(x = team, y = win_prob, fill = team)) +
  geom_col() +
  geom_label(aes(label = win_prob, accuracy = 0.1),color = "black", fill = "white") +
  scale_fill_manual(values = c("PHI" = "#004C54", "SF" = "gold")) +
  labs(title = "Wild Card Predictions",
       y = "Win Probability",
       x = "Teams")
```

```
## Warning in geom_label(aes(label = win_prob, accuracy = 0.1), color = "black", :
## Ignoring unknown aesthetics: accuracy
```

```
nfc_wc2_chart
```

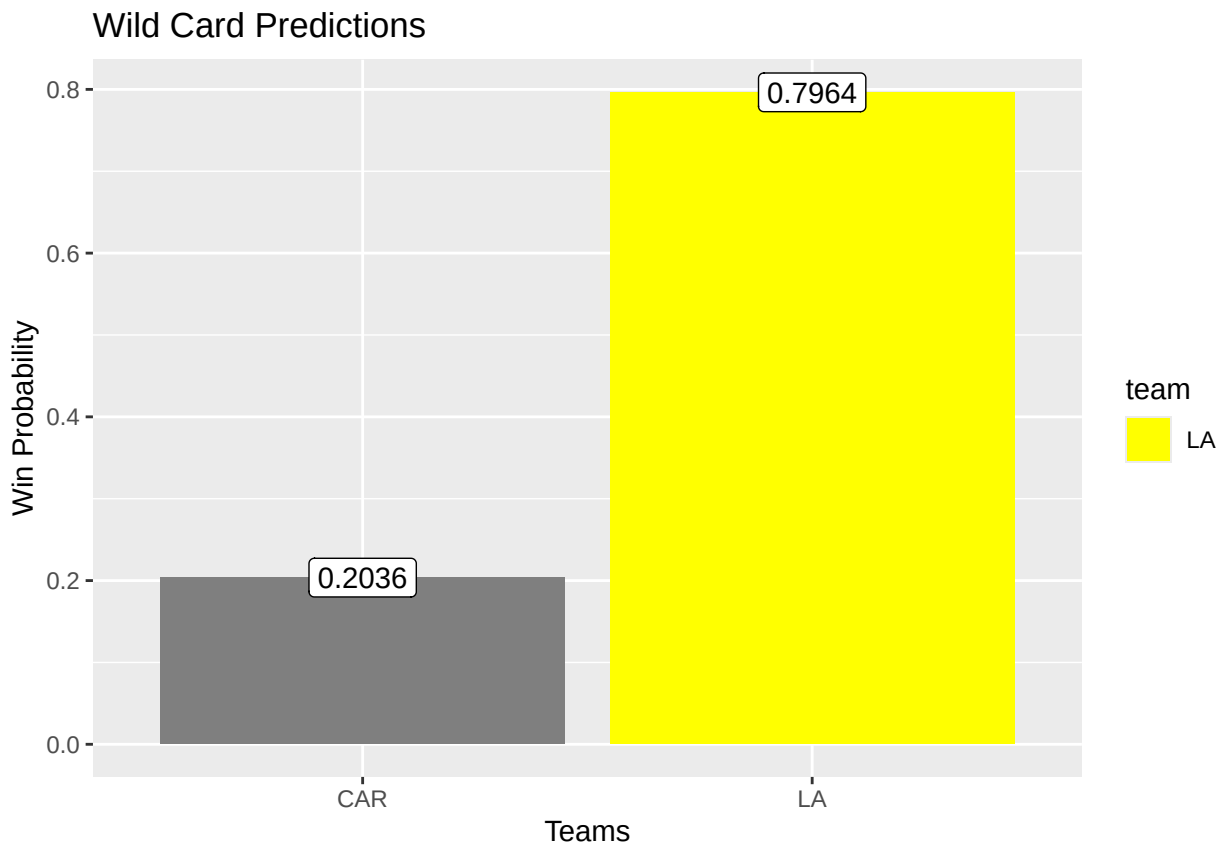



```
LA_CAR <- wc_odds %>%
  filter(team %in% c("LA", "CAR"))

nfc_wc3_chart <- ggplot(LA_CAR, aes(x = team, y = win_prob, fill = team)) +
  geom_col() +
  geom_label(aes(label = win_prob, accuracy = 0.1),color = "black", fill = "white") +
  scale_fill_manual(values = c("LA" = "yellow", "Car" = "steelblue")) +
  labs(title = "Wild Card Predictions",
       y = "Win Probability",
       x = "Teams")
```

```
## Warning in geom_label(aes(label = win_prob, accuracy = 0.1), color = "black", :
## Ignoring unknown aesthetics: accuracy
```

```
nfc_wc3_chart
```

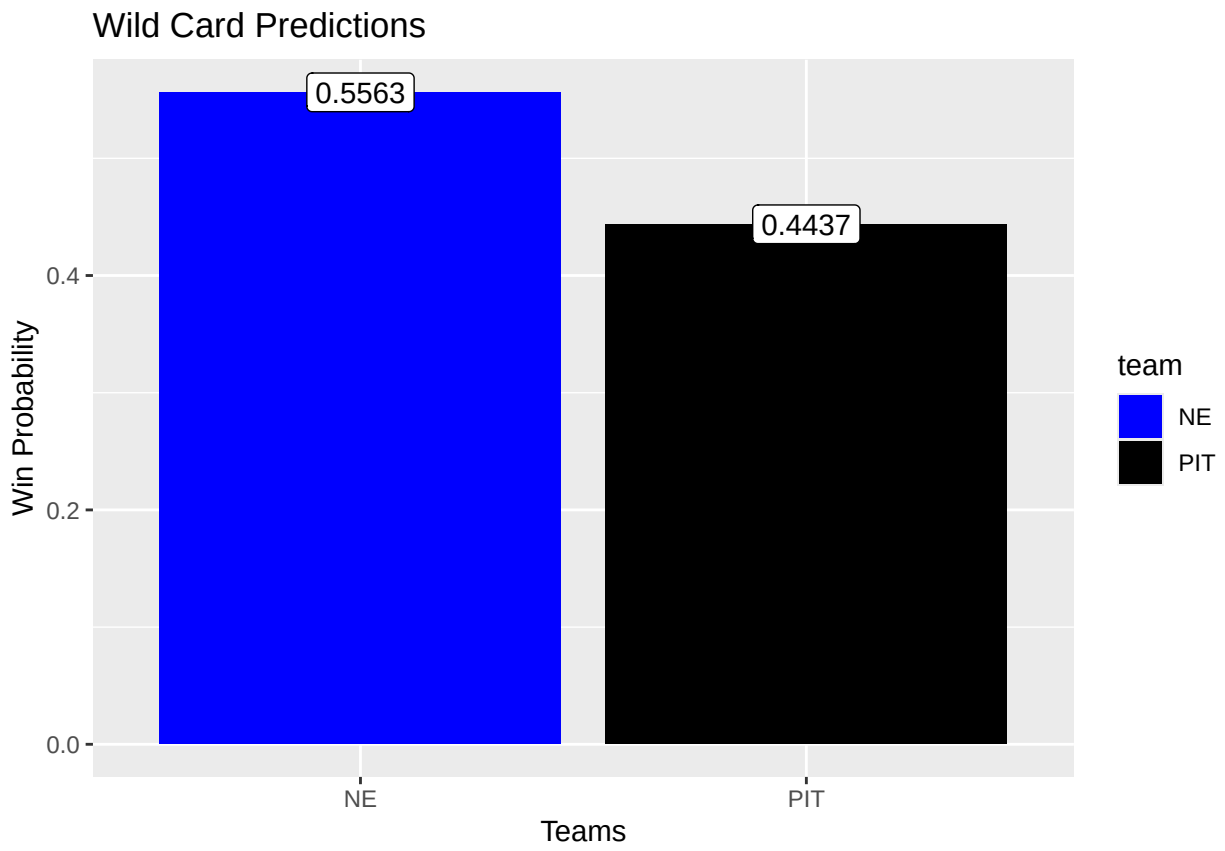



```
NE_PIT <- dv_odds %>%
  filter(team %in% c("NE", "PIT"))

afc_dv1_chart <- ggplot(NE_PIT, aes(x = team, y = win_prob, fill = team)) +
  geom_col() +
  geom_label(aes(label = win_prob, accuracy = 0.1),color = "black", fill = "white") +
  scale_fill_manual(values = c("NE" = "blue", "PIT" = "black")) +
  labs(title = "Wild Card Predictions",
       y = "Win Probability",
       x = "Teams")
```

```
## Warning in geom_label(aes(label = win_prob, accuracy = 0.1), color = "black", :
## Ignoring unknown aesthetics: accuracy
```

```
afc_dv1_chart
```

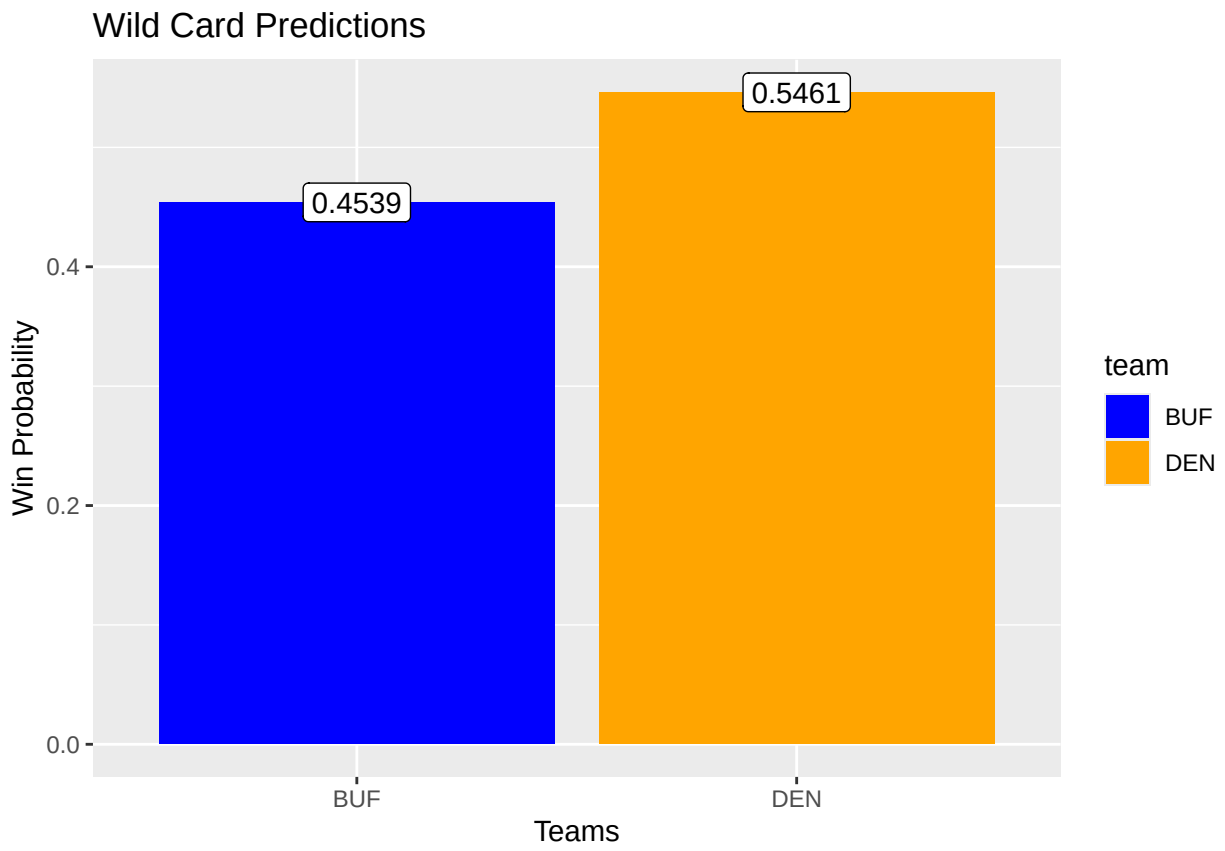



```
DEN_BUF <- dv_odds %>%
  filter(team %in% c("DEN", "BUF"))

afc_dv2_chart <- ggplot(DEN_BUF, aes(x = team, y = win_prob, fill = team)) +
  geom_col() +
  geom_label(aes(label = win_prob, accuracy = 0.1),color = "black", fill = "white") +
  scale_fill_manual(values = c("DEN" = "orange", "BUF" = "blue")) +
  labs(title = "Wild Card Predictions",
       y = "Win Probability",
       x = "Teams")
```

```
## Warning in geom_label(aes(label = win_prob, accuracy = 0.1), color = "black", :
## Ignoring unknown aesthetics: accuracy
```

```
afc_dv2_chart
```

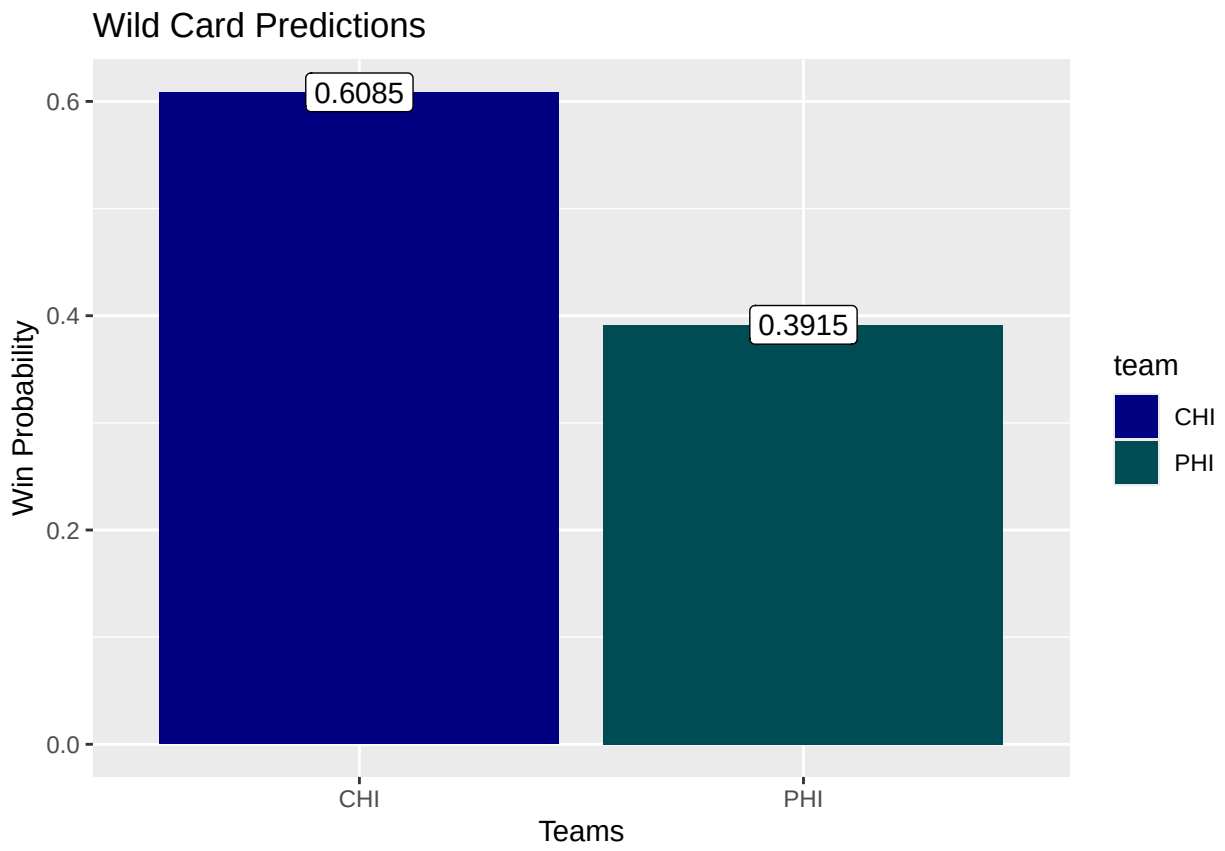



```
CHI_PHI <- dv_odds %>%
  filter(team %in% c("CHI", "PHI"))

nfc_dv1_chart <- ggplot(CHI_PHI, aes(x = team, y = win_prob, fill = team)) +
  geom_col() +
  geom_label(aes(label = win_prob, accuracy = 0.1),color = "black", fill = "white") +
  scale_fill_manual(values = c("CHI" = "navy", "PHI" = "#004C54")) +
  labs(title = "Wild Card Predictions",
       y = "Win Probability",
       x = "Teams")
```

```
## Warning in geom_label(aes(label = win_prob, accuracy = 0.1), color = "black", :
## Ignoring unknown aesthetics: accuracy
```

```
nfc_dv1_chart
```

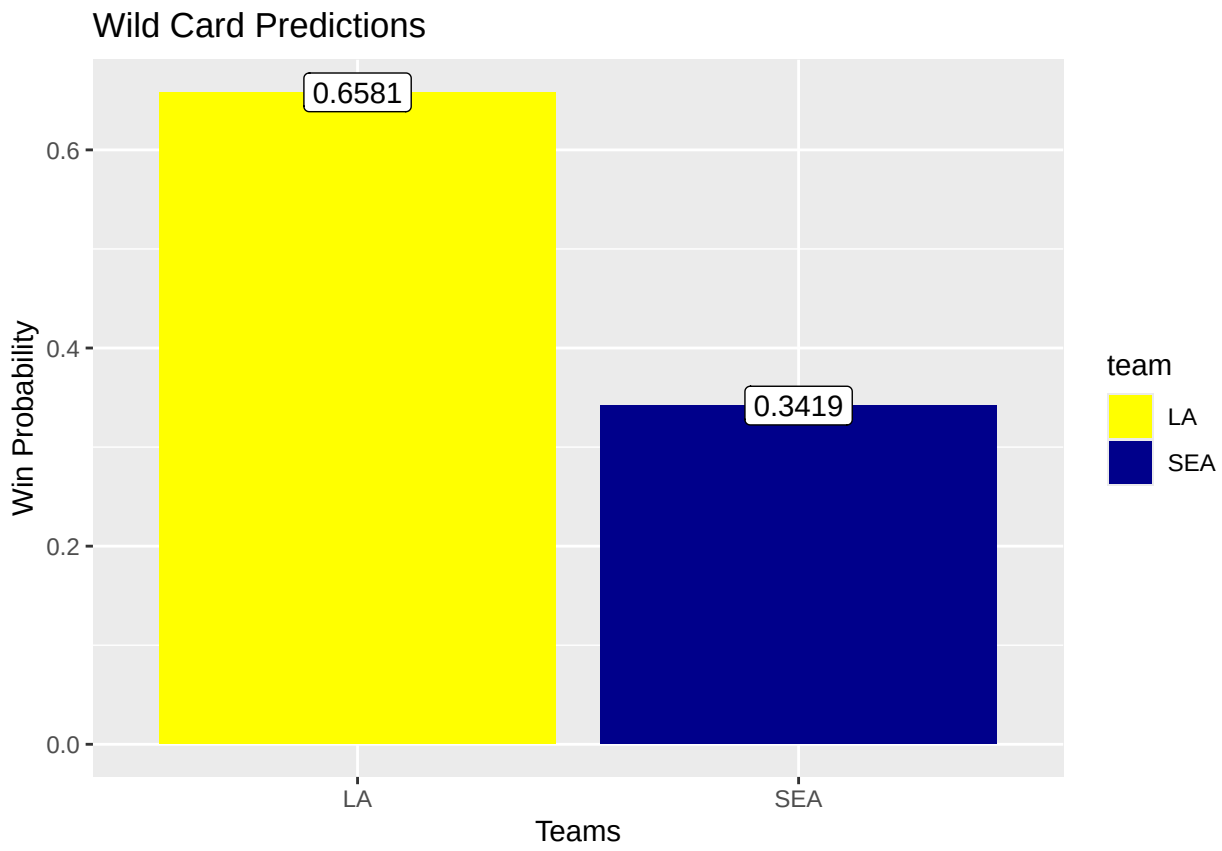



```
LA_SEA <- dv_odds %>%
  filter(team %in% c("LA", "SEA"))

nfc_dv2_chart <- ggplot(LA_SEA, aes(x = team, y = win_prob, fill = team)) +
  geom_col() +
  geom_label(aes(label = win_prob, accuracy = 0.1),color = "black", fill = "white") +
  scale_fill_manual(values = c("LA" = "yellow", "SEA" = "darkblue")) +
  labs(title = "Wild Card Predictions",
       y = "Win Probability",
       x = "Teams")
```

```
## Warning in geom_label(aes(label = win_prob, accuracy = 0.1), color = "black", :
## Ignoring unknown aesthetics: accuracy
```

```
nfc_dv2_chart
```

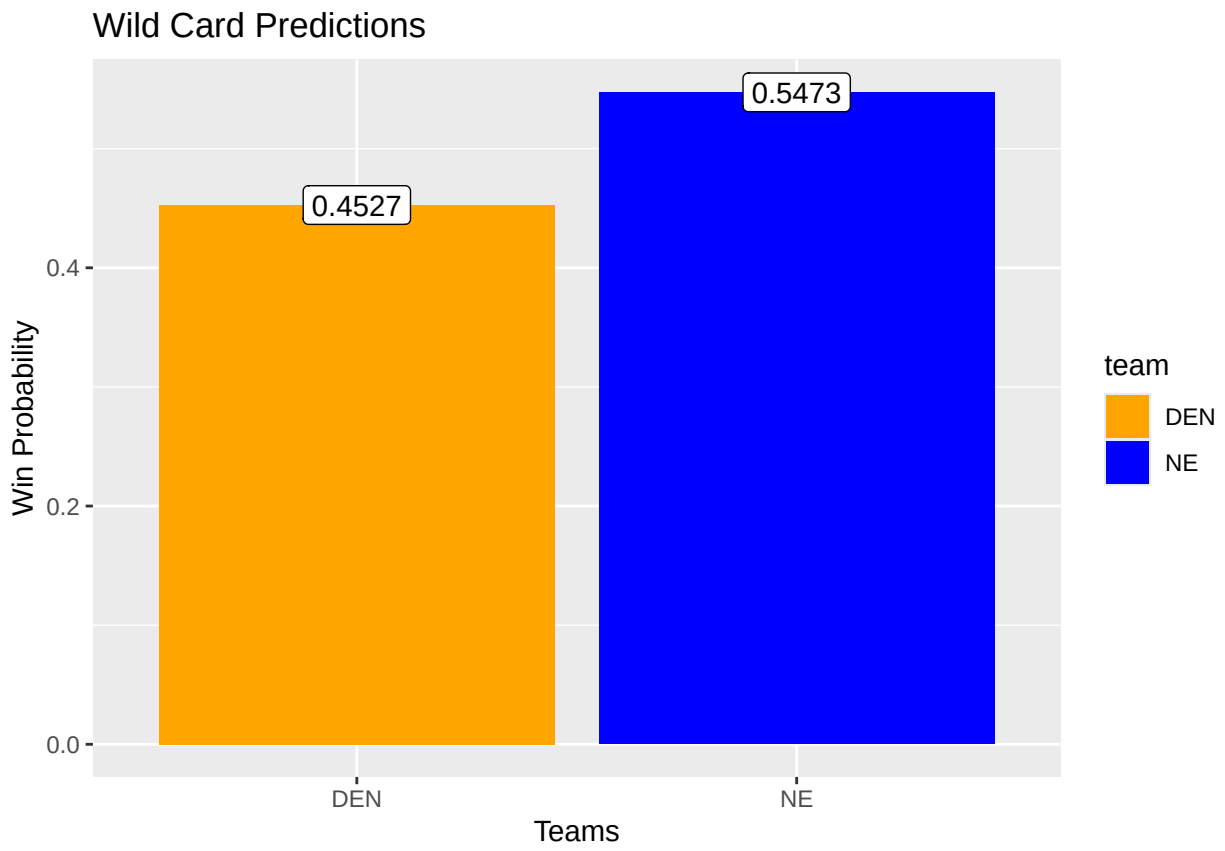



```
DEN_NE <- ch_odds %>%
  filter(team %in% c("DEN", "NE"))

afc_ch_chart <- ggplot(DEN_NE, aes(x = team, y = win_prob, fill = team)) +
  geom_col() +
  geom_label(aes(label = win_prob, accuracy = 0.1),color = "black", fill = "white") +
  scale_fill_manual(values = c("DEN" = "orange", "NE" = "blue")) +
  labs(title = "Wild Card Predictions",
       y = "Win Probability",
       x = "Teams")
```

```
## Warning in geom_label(aes(label = win_prob, accuracy = 0.1), color = "black", :
## Ignoring unknown aesthetics: accuracy
```

```
afc_ch_chart
```

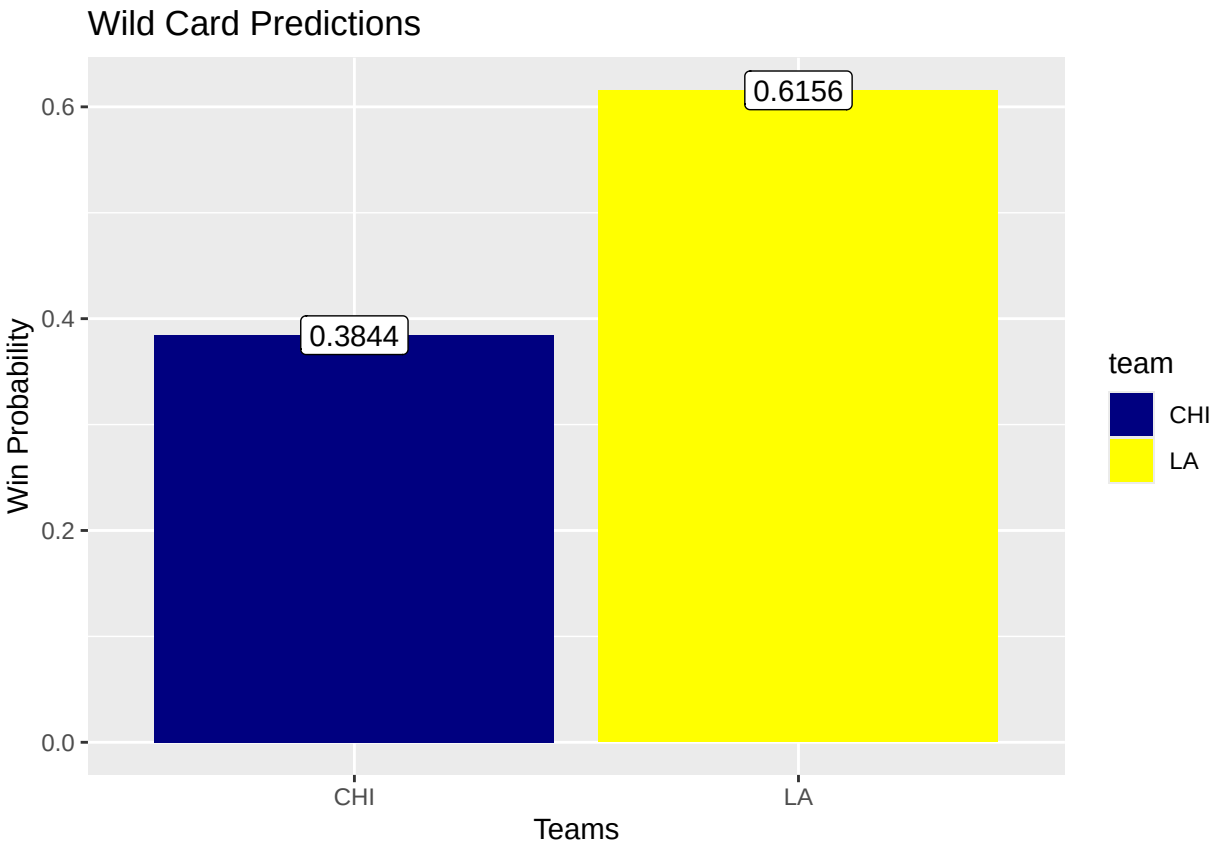



```
LA_CHI <- ch_odds %>%
  filter(team %in% c("LA", "CHI"))

nfc_ch_chart <- ggplot(LA_CHI, aes(x = team, y = win_prob, fill = team)) +
  geom_col() +
  geom_label(aes(label = win_prob, accuracy = 0.1),color = "black", fill = "white") +
  scale_fill_manual(values = c("LA" = "yellow", "CHI" = "navy")) +
  labs(title = "Wild Card Predictions",
       y = "Win Probability",
       x = "Teams")
```

```
## Warning in geom_label(aes(label = win_prob, accuracy = 0.1), color = "black", :
## Ignoring unknown aesthetics: accuracy
```

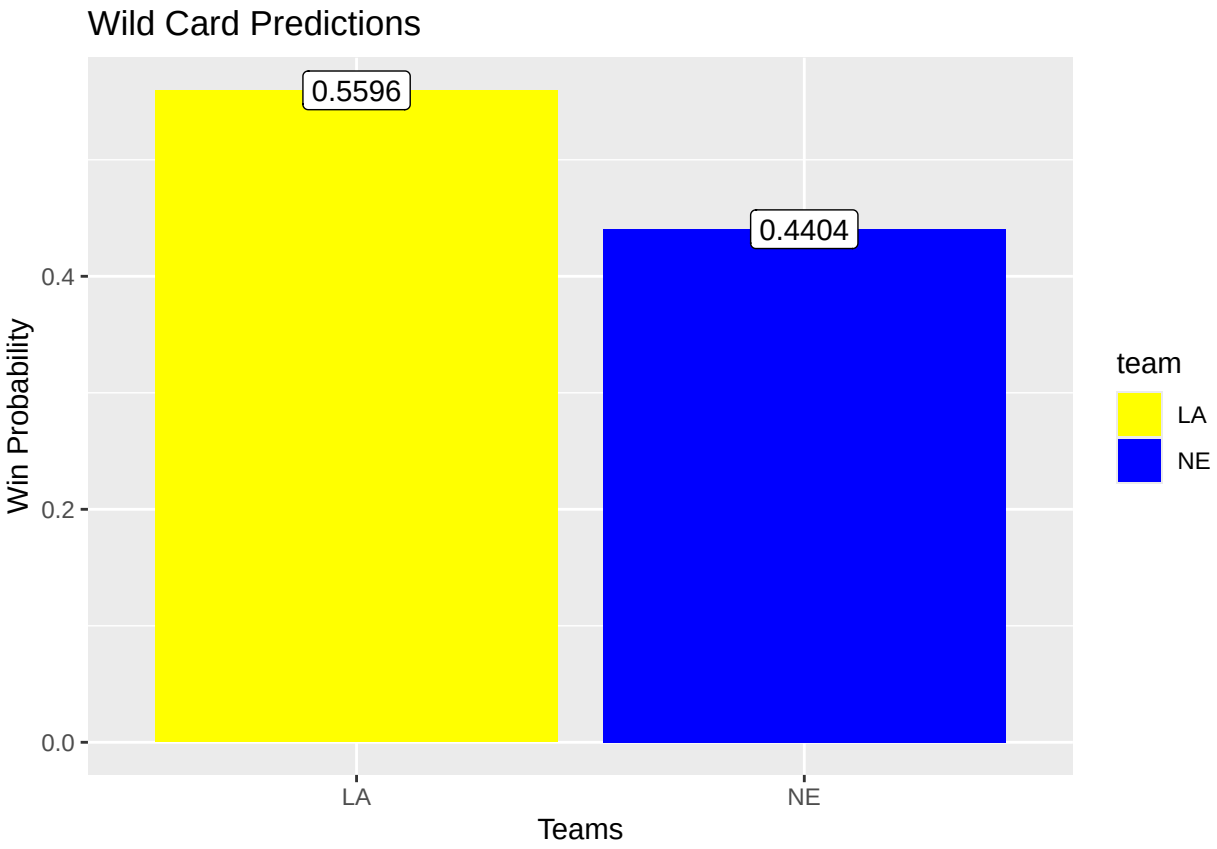
```
nfc_ch_chart
```




```
sb_chart <- ggplot(sb_odds, aes(x = team, y = win_prob, fill = team)) +
  geom_col() +
  geom_label(aes(label = win_prob, accuracy = 0.1),color = "black", fill = "white") +
  scale_fill_manual(values = c("LA" = "yellow", "NE" = "blue")) +
  labs(title = "Wild Card Predictions",
       y = "Win Probability",
       x = "Teams")
```

```
## Warning in geom_label(aes(label = win_prob, accuracy = 0.1), color = "black", :
## Ignoring unknown aesthetics: accuracy
```

```
sb_chart
```


Wild Card Predictions
0.5596
0.4404
0.4
0.2
0.0
Win Probability
LA
NE
Teams
team
LA
NE